

CSE 4417 IoT Theory

Lecture 06: Controllers (Arduino)

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Computer Basics

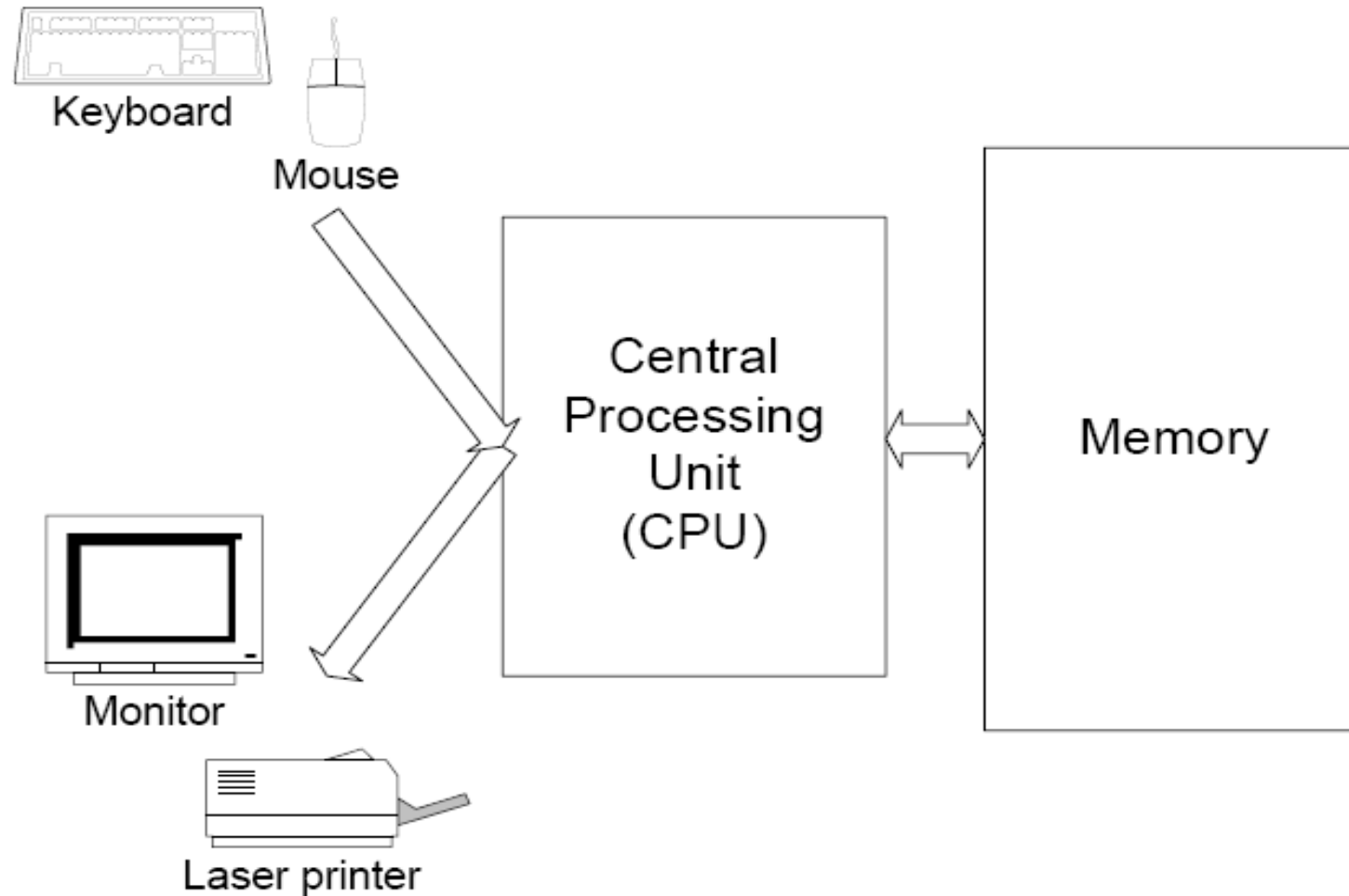
The word "Computer" is derived from the term "compute," which means to calculate.

- A computer is an electronic machine that can:
- Store, retrieve, and process (manipulate/convert) data.
- Perform computations and make logical decisions.

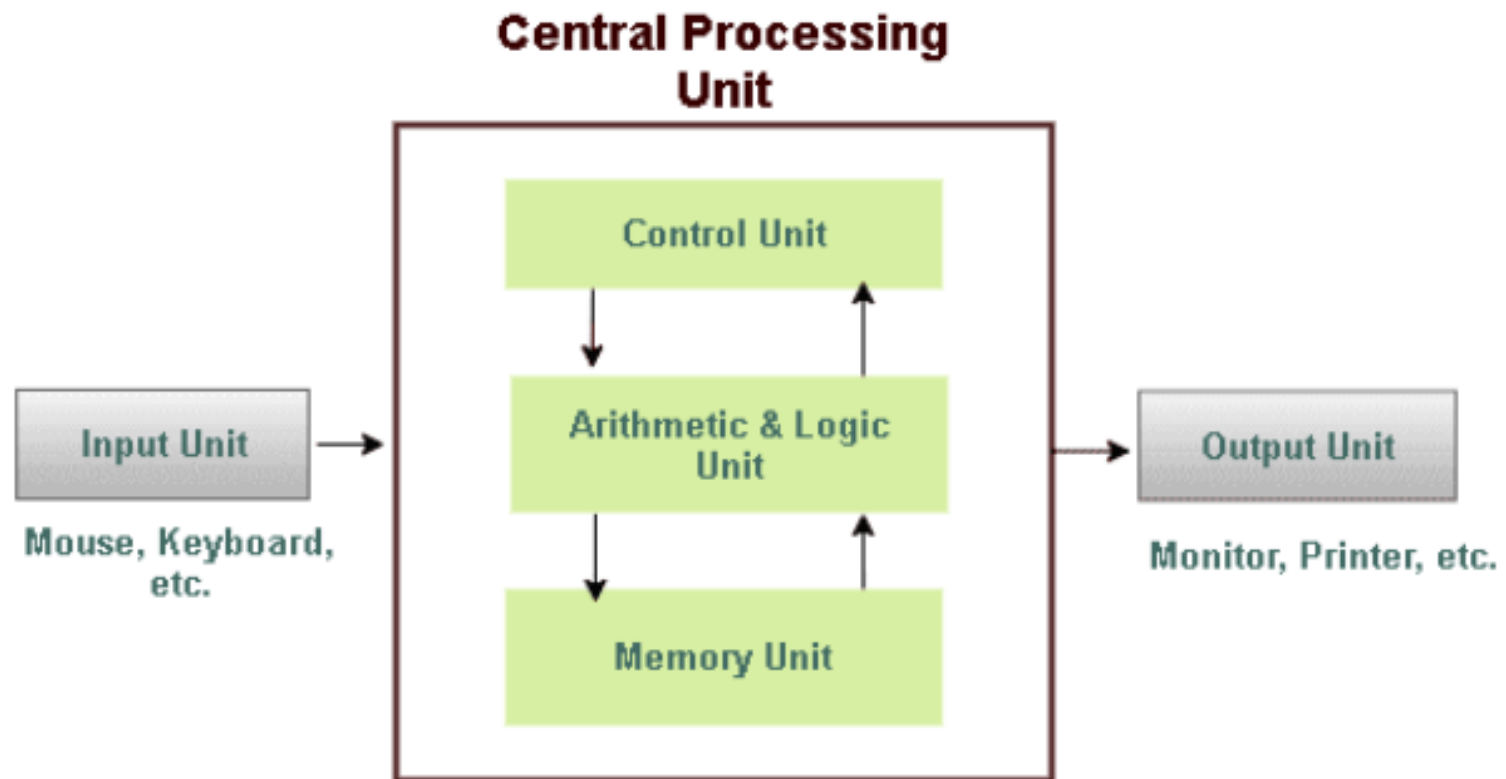
Key Characteristics:

- It has no intelligence or power of its own.
- It cannot think or act independently.

A Typical Computer



A Typical Computer



Work flow of Computer

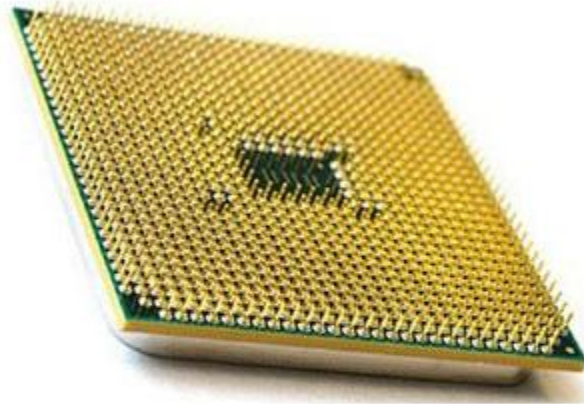


- Input data from disk, keyboard, mouse or from other storage media
- Process data (Processing units are CPU and Main Memory)
- Output results to display screen or to store in hard disk or other media

Introduction to Microprocessors and Microcontrollers

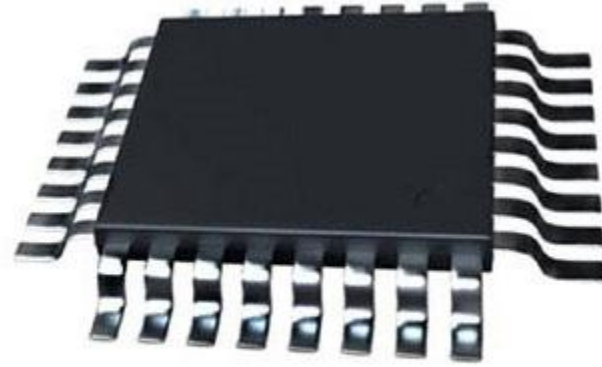
- Microprocessor: A central processing unit (CPU) on a single chip, primarily for computing tasks.
- Microcontroller: An integrated chip with a CPU, memory, and peripherals designed for embedded applications.

Introduction to Microprocessors and Microcontrollers



Microprocessor

V/S



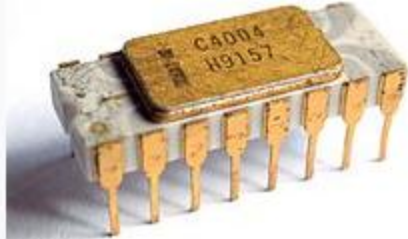
Microcontroller

Microprocessors

- CPU on a single chip, the brain of a computing system.
- Handles general-purpose tasks: running software, calculations, and operations.
- Used in PCs, servers, and gaming consoles for high computational power.
- Example- Intel Core i7, AMD Ryzen 7, ARM Cortex-A series, Apple M1/M2, Qualcomm Snapdragon.

Microprocessors

Intel 4004



Intel C4004 processor with grey traces

General information

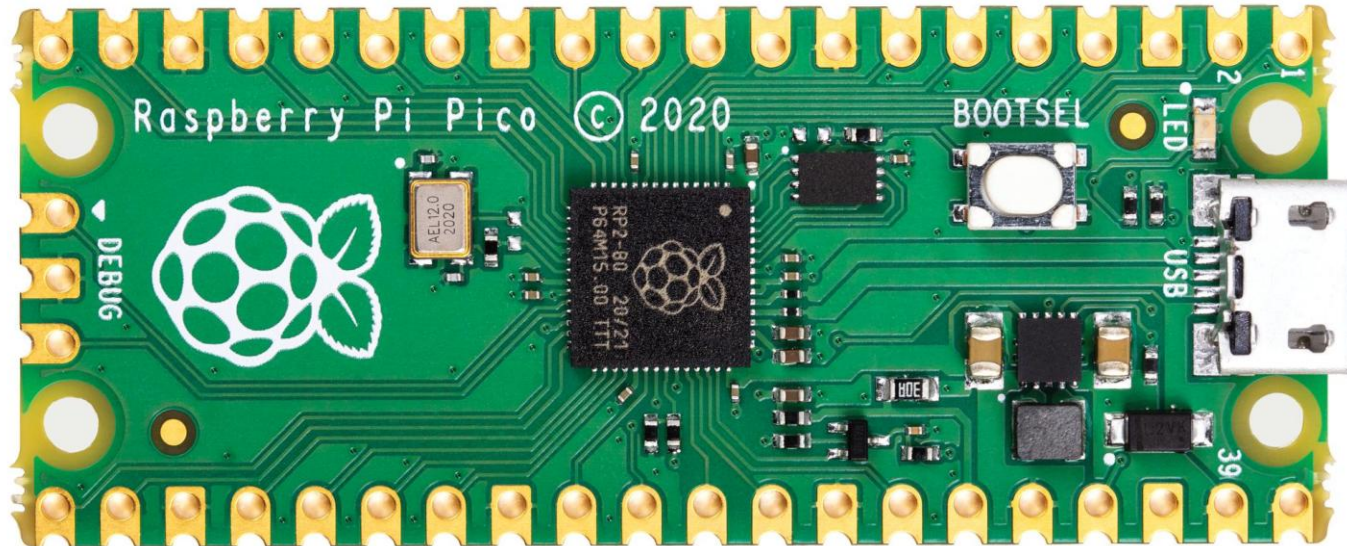
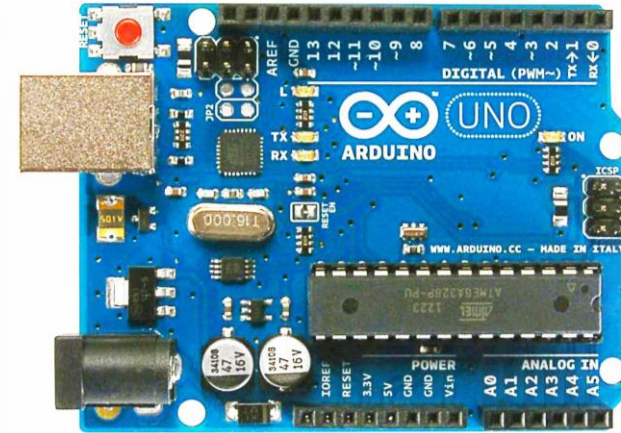


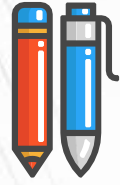
© www.cpu-world.com

Introduction to Microcontroller

- A complete system on a chip.
- Integrates:
 - CPU for processing.
 - Memory (RAM for data, ROM/Flash for program storage).
 - Peripherals (I/O ports, timers, ADC, UART, SPI, I2C)
- Designed for specific embedded applications.
- Low power consumption and compact size.
- Examples- Arduino, ESP32, PIC16, ATmega328, STM32.

Introduction to Microcontroller



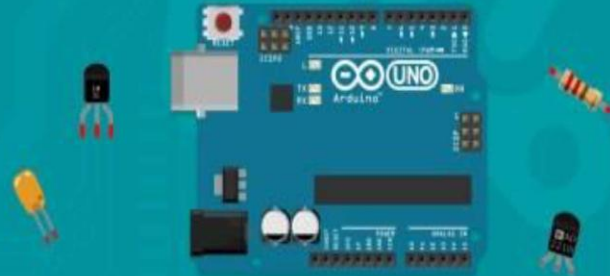


What is Arduino?

Welcome To Arduino

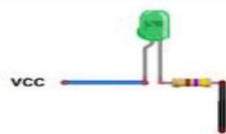
Apprendre

 **ARDUINO**





WELCOME To Arduino



LED Blinking



LED Blinking



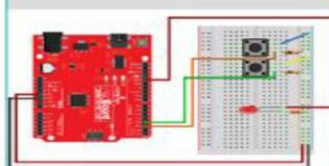
Arduino software



Arduino Hardware



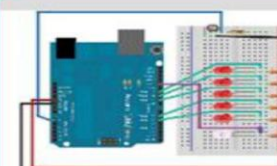
Key and LED



Push bottom



Push bottom



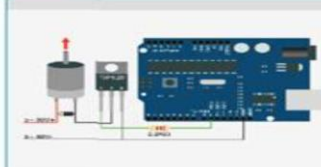
Array LED Blinking



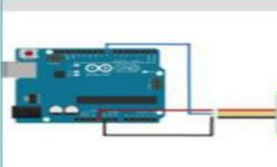
Serial communication



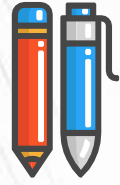
Analog reading



PWM



Sensor introduction



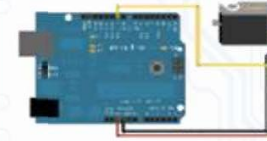
WELCOME To Arduino



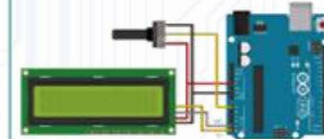
Photo cell
Sensor introduction



Ultrasonic
Sensor introduction



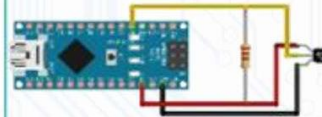
Servo motor



LCD



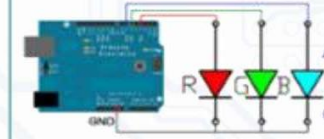
7segment
সাত সেকশন বিশিষ্ট



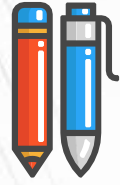
Ds18b20 sensor
তাপমাত্রা সেন্সর



DHT temp sensor
আর্ডুইনো ডিএইচটি



RGB LED



What is Arduino?

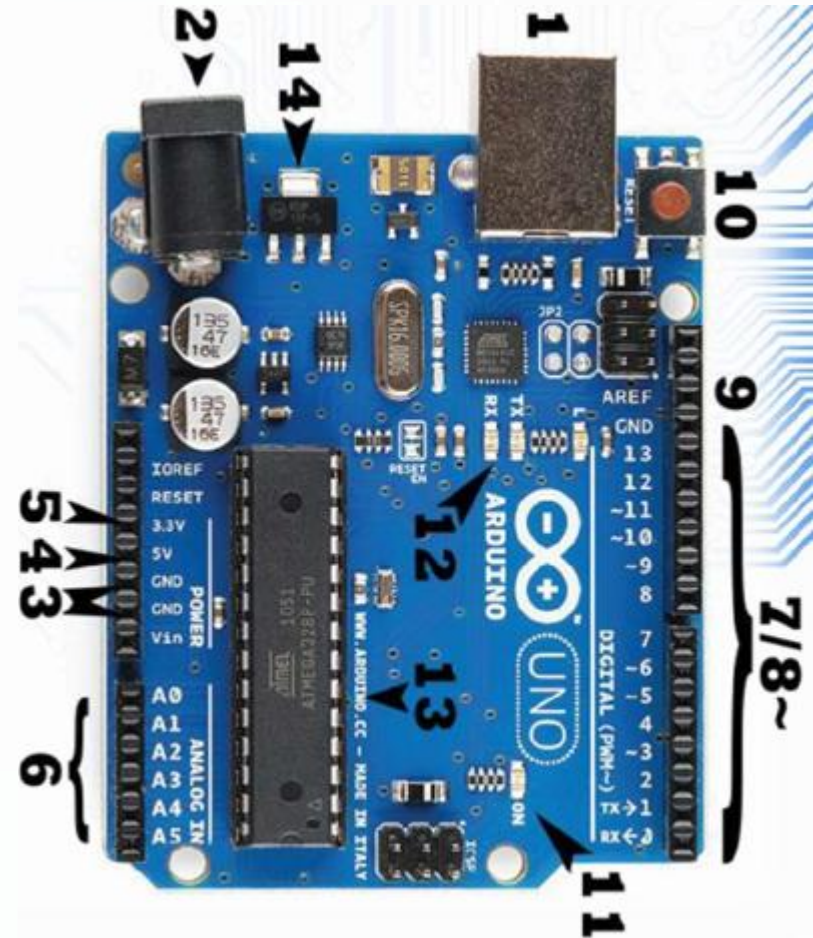
- Arduino is an open-source prototyping platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.
- You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.



Arduino PINs

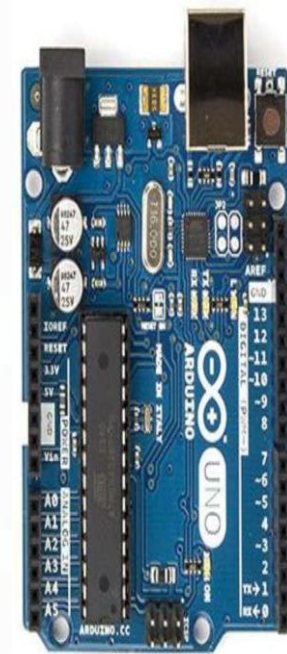
Most of this part are
common in Arduino
boards

- (1) ,(2) Power (USB / Barrel Jack)
- Pins (5V, 3.3V, GND, Analog, Digital, PWM, AREF)
(3) GND
(4) 5V & (5) 3.3V
- (6)Analog
- (7)Digital
- (8) **PWM** : (~) next to some of the
digital pins (3, 5, 6, 9, 10, and 11 on the UNO).
- (8) **AREF** :
- (9) **Reset Button**
- (10) **Power LED Indicator**
- (11) **TX RX LEDs**
- (12) **Main IC** (14) **Voltage Regulator**



Step 2: Before you begin...

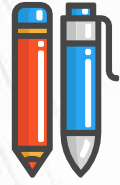
- Before you begin plugging things into your new Arduino, it may be good to know what can damage the board.
 - Drawing more than 40mA from an output pin.
 - An Arduino can only supply 40mA per output pin, so you cannot drive a motor or a speaker directly, for example, and you cannot connect an LED directly (without a resistor).
 - Shorting an output in to the +5v, +3.3v or the ground pins, will also kill your board: If an output pin is at 5v for example, and you connect it to the ground, it draws an enormous amount of current, and kills your Arduino almost instantly.
 - The pins go through the circuit board, so make sure you don't place the Arduino on a conductive (metal) surface, because it will short out the pins.



Step 2: Before you begin...

- Drawing more than 200mA from all output pins together.
The ATmega chip on your Arduino can only supply 200mA in total, so driving more than 10 LEDs @ 20mA each, for example, will eventually damage your board.
- Supplying more than 5v (3.3v) to an input pin.
Supplying more than the operating voltage of the Arduino on any pin is very dangerous.
- Some Arduinos that run at 3.3v have 5v tolerant pins, but that's about it.
- **This also holds true for other devices, like sensors or wireless chips: always check the voltages: if you connect the output of a 5V Arduino to a 3.3V chip, you might kill it.**

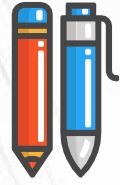




Step 2: Before you begin...

- **Supplying more than 5v to the 5v pin.**
The 5v of the Arduino board goes directly to the ATmega chip, that is rated for an absolute maximum of 6v.
- Supplying more than 12v to the Vin pin.
There's an onboard 5v voltage regulator on the board, that will overheat and die if you feed it with more than 12v.
- Drawing more than 500mA from the 5v pin (when running off an external power supply).
The onboard 5v voltage regulator can only supply 500mA of current. The 5vUSB has a polyfuse to limit the current to 500mA.

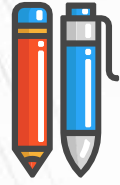




Step 2: Before you begin...

- **Drawing more than 50mA from the 3.3v pin.**
The onboard 3.3v voltage regulator can only supply 50mA of current. This means that you can not connect power hungry 3.3v devices like an ESP8266 or nRF24L01 directly to the Arduino: you need an external 3.3v voltage regulator.
- **Reversing the polarity of the power supply.**
If you swap the 5v or Vin pin with the GND pin, you'll kill the board almost instantly. The barrel jack has a diode to protect against reverse polarity.
- **Connecting a load to the Vin pin while using USB power.**
If you connect a load to the Vin pin while the 5v to the Arduino comes from the USB connection, current will flow backwards through the voltage regulator, damaging it.
- **Static electricity**
Although most chips have clamping diodes as protection against ESDs (electrostatic discharges), it may be wise to use an anti-static wrist strap, or to remove the carpet under your desk.





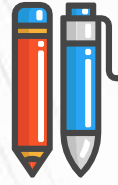
Arduino Types

- Many different versions
 - Number of input/output channels
 - Form factor
 - Processor
- Leonardo
- Due
- Micro
- LilyPad
- Esplora
- Uno



Arduino provides many alternative boards.

- These boards vary in the following aspects:
- the microcontroller type,
- The microcontroller speed (frequency)
- the physical size
- the number of input and output pins
- the memory space for programs, and
- the board price.



Arduino Types



Arduino UNO R3
atmega328pwith USB cable



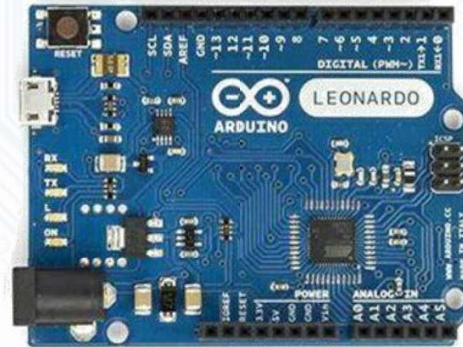
Arduino UNO R3
mega328pwith USB cable



Arduino Pro Mini
atmega3285V/16M



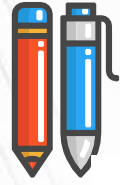
Arduino MEGA2560 R3
withUSB cable(With logo)



Arduino Leonardo
R3ATMEGA32U4
with USB cable

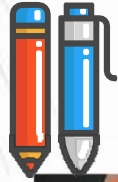


Arduino Nano
V3.0with USB
cable



Arduino Types





Arduino Types



Arduino RS232
(male pins)



Arduino
Diecimila



Arduino
Duemilanove
(rev 2009b)



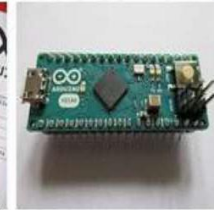
Arduino Uno R2



Arduino Uno
SMD R3



Arduino Leonardo



Arduino
micro (AtMega
32U4)



Arduino pro micro
(AtMega32U4)



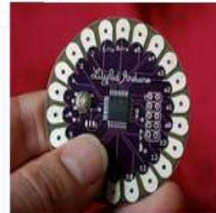
Arduino Pro
(No USB)



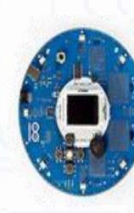
Arduino Mega



Arduino
Nano
(DIP-30
footprint)



Arduino LilyPad
00_ (rev 2007)
(No USB)



Arduino Robot



Arduino Esplora



Arduino Ethernet
(AVR + W5100)



Arduino Yun
(AVR + AR9331)

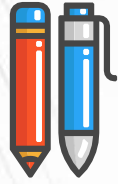


Arduino Due
(ARM Cortex-M3 core)

UNO

- The Uno is a great choice for your first Arduino. It's got everything you need to get started, and nothing you don't. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a USB connection, a power jack, a reset button and more. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.
- The pins are in three groups:
 -
 - **Invented in 2010**
 - **14 digital pins**
 - **6 analog pins**
 - **power**





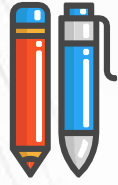
Leonardo

The Leonardo is Arduino's first development board to use one microcontroller with built-in USB. This means that it can be cheaper and simpler. Also, because the board is handling USB directly, code libraries are available which allow the board to emulate a computer keyboard, mouse, and more!

- Compared to the Uno, a slight
- upgrade. Built in USB compatibility
- Bugs?

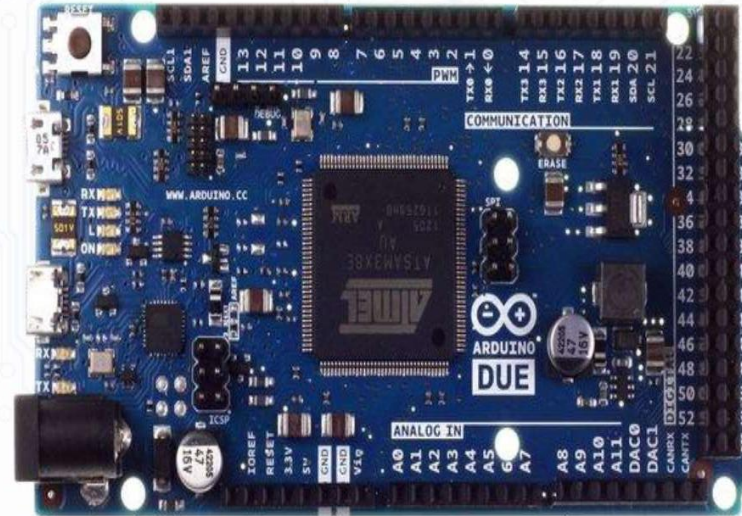
■ Presents to PC as a mouse or keyboard



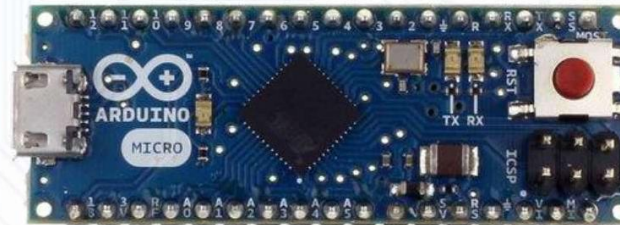


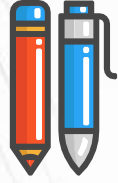
Due

- **Much faster processor, many more pins**
- **Operates on 3.3 volts**
- **Similar to the Mega**



When size matters: Micro, Nano, Mini
Includes all functionality of the
Leonardo Easily usable on a breadboard

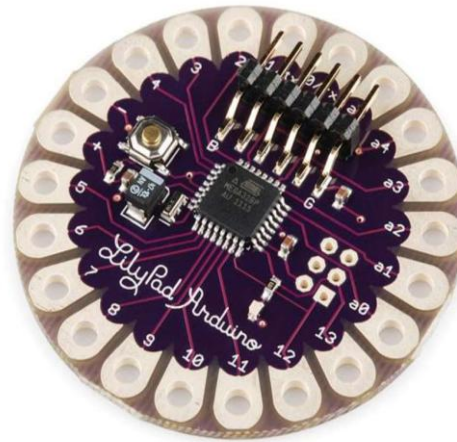




LilyPad

This is LilyPad Arduino main board! LilyPad is a wearable e-textile technology developed by Leah Buechley and cooperatively designed by Leah and SparkFun. Each LilyPad was creatively designed with large connecting pads and a flat back to allow them to be sewn into clothing with conductive thread. The LilyPad also has its own family of input, output, power, and sensor boards that are also built specifically for e-textiles. They're even washable!

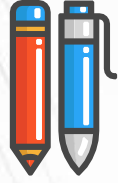
- **LilyPad is popular for clothing-based projects.**



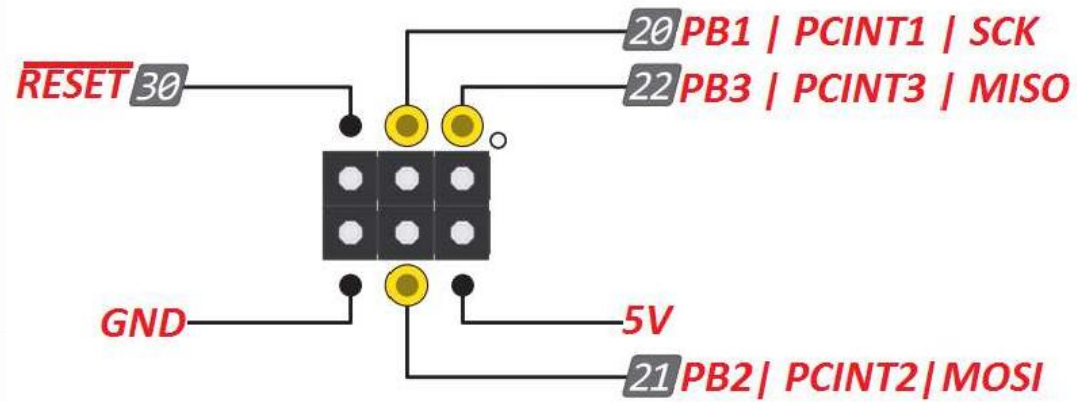








Arduino ICSP Pinout



In Circuit Serial Programming Pinout (ICSP)

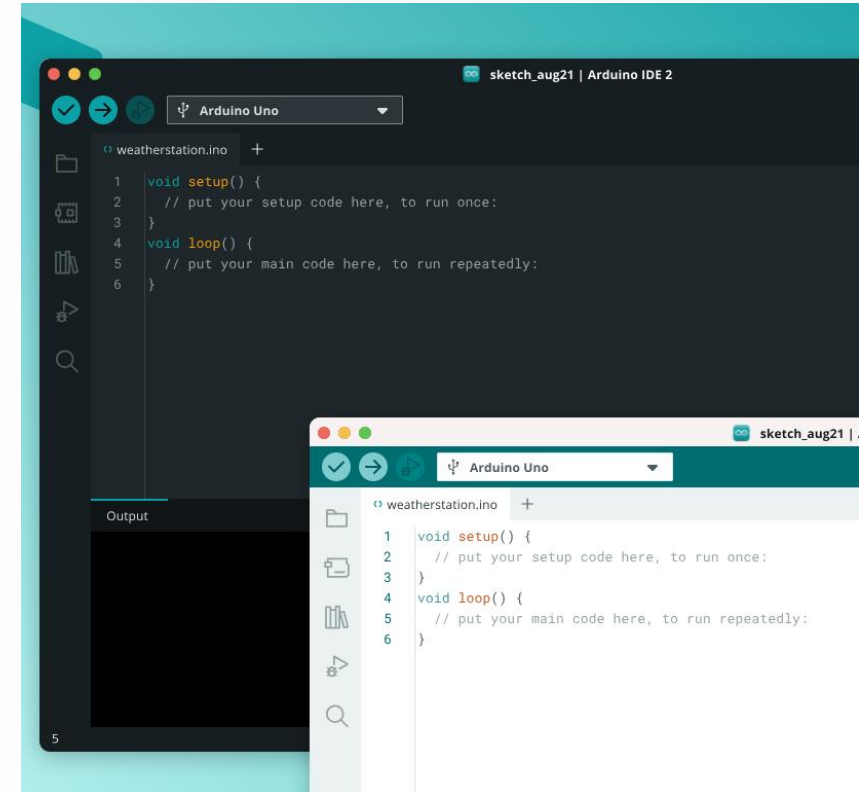
www.CircuitsToday.com

Arduino Types

NO	Board name	PROCESSOR	Voltage	Crystal
1	Arduino Uno	ATmega328	5 V/7-12 V	16MHz
2	Due	AT91SAM3X8E	3.3 V/7-12 V	84 MHz
3	Leonardo	ATmega32u4	5 V/7-12 V	16MHz
4	Mega 2560	ATmega2560	5 V/7-12 V	16MHz
5	Mega ADK	ATmega2560	5 V/7-12 V	16MHz
6	Micro	ATmega32u4	5 V/7-12 V	16MHz
7	Mini	ATmega328	5 V/7-9 V	16MHz
8	Nano	ATmega168 ATmega328	5 V/7-9 V	16MHz
9	Ethernet	ATmega328	5 V/7-12 V	16MHz
10	Esplora	ATmega32u4	5 V/7-12 V	16MHz
11	ArduinoBT	ATmega328	5 V/2.5-12 V	16MHz
12	Fio	ATmega328P	3.3 V/3.7-7 V	8MHz
13	Pro (168)	ATmega168	3.3 V/3.35-12 V	8MHz
14	Pro (328)	ATmega328	5 V/7-12 V	16MHz
15	Pro Mini	ATmega168	3.3 V/3.35-12 V 5 V/5-12 V	8MHz 16MHz
16	LilyPad	ATmega168V ATmega328V	2.7-5.5 V/2.7-5.5 V	8MHz
17	LilyPad USB	ATmega32u4	3.3 V/3.8-5V	8MHz
18	LilyPad Simple	ATmega328	2.7-5.5 V/2.7-5.5 V	8MHz
19	LilyPad SimpleSnap	ATmega328	2.7-5.5 V/2.7-5.5 V	8MHz
20	Yun	ATmega32u4	5 V	16MHz

Arduino Compiler

- Download current compiler from: arduino.cc/en/Main/software
- **Arrogantly refers to itself as an IDE .**
- **Run the software installer.**
- **Written in Java, it is fairly slow.**



Visit [playground.arduino.cc/Main/ DevelopmentTools](https://playground.arduino.cc/Main/DevelopmentTools)
for alternatives to the base arduino IDE

Arduino Program Development

- Based on C++ without 80% of the instructions. A handful of new commands.
- Programs are called
- 'sketches'. Sketches need two
- functions:
 - `void setup( )`
 - `void loop( )`
 - `setup( )` runs first and once.
 - `loop( )` runs over and over, until power is lost or a new sketch is loaded.

Arduino C Specific Functions

- **pinMode(pin,mode)**
 - Designates the specified pin for input or output
- **digitalWrite(pin, value)**
 - Sends a voltage level to the designated pin
- **digitalRead(pin)**
 - Reads the current voltage level from the designated pin
- **analog versions of above**
 - analogRead's range is 0 to 1023
- **serial commands**
 - print, println, write

How to Setup Arduino IDE on your Local System



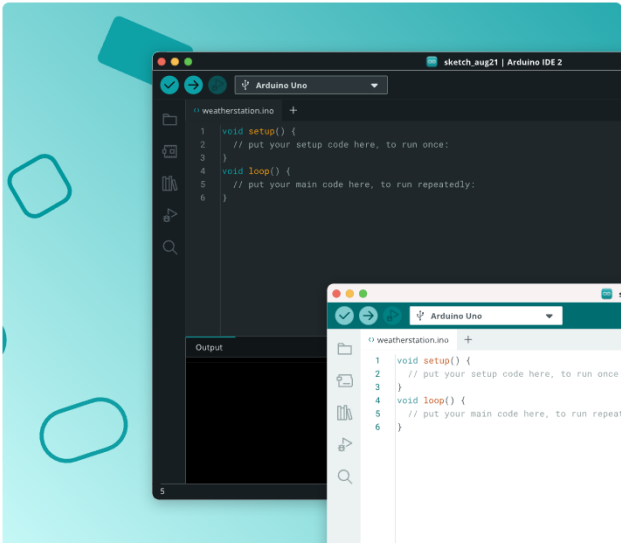
For Professionals For Education For Makers

Products ▾ Community ▾ Documentation ▾

SHOP

Cloud

Bring Your Projects to Life with Arduino Software



**Arduino IDE 2.3.6**
[Release notes](#)

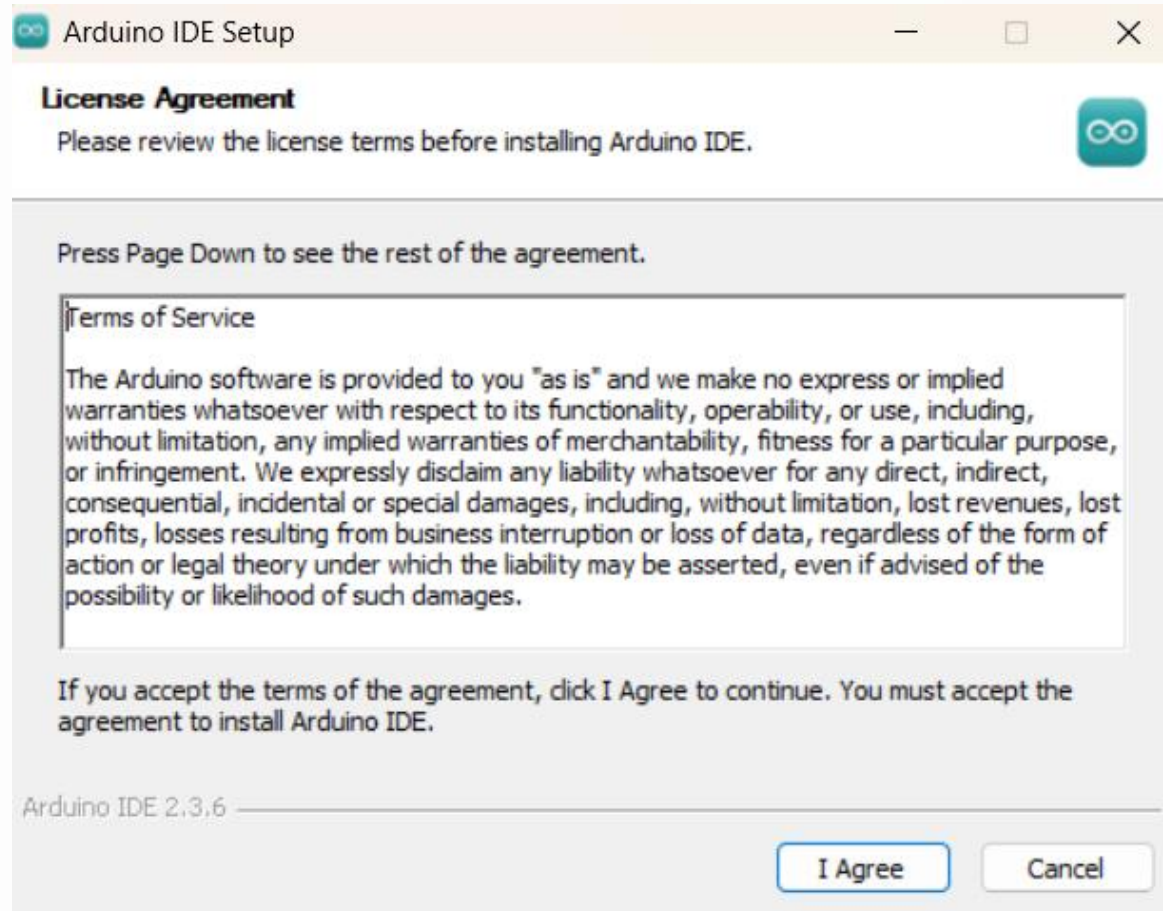
The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).

Windows Win 10 or newer (64-bit) **DOWNLOAD**

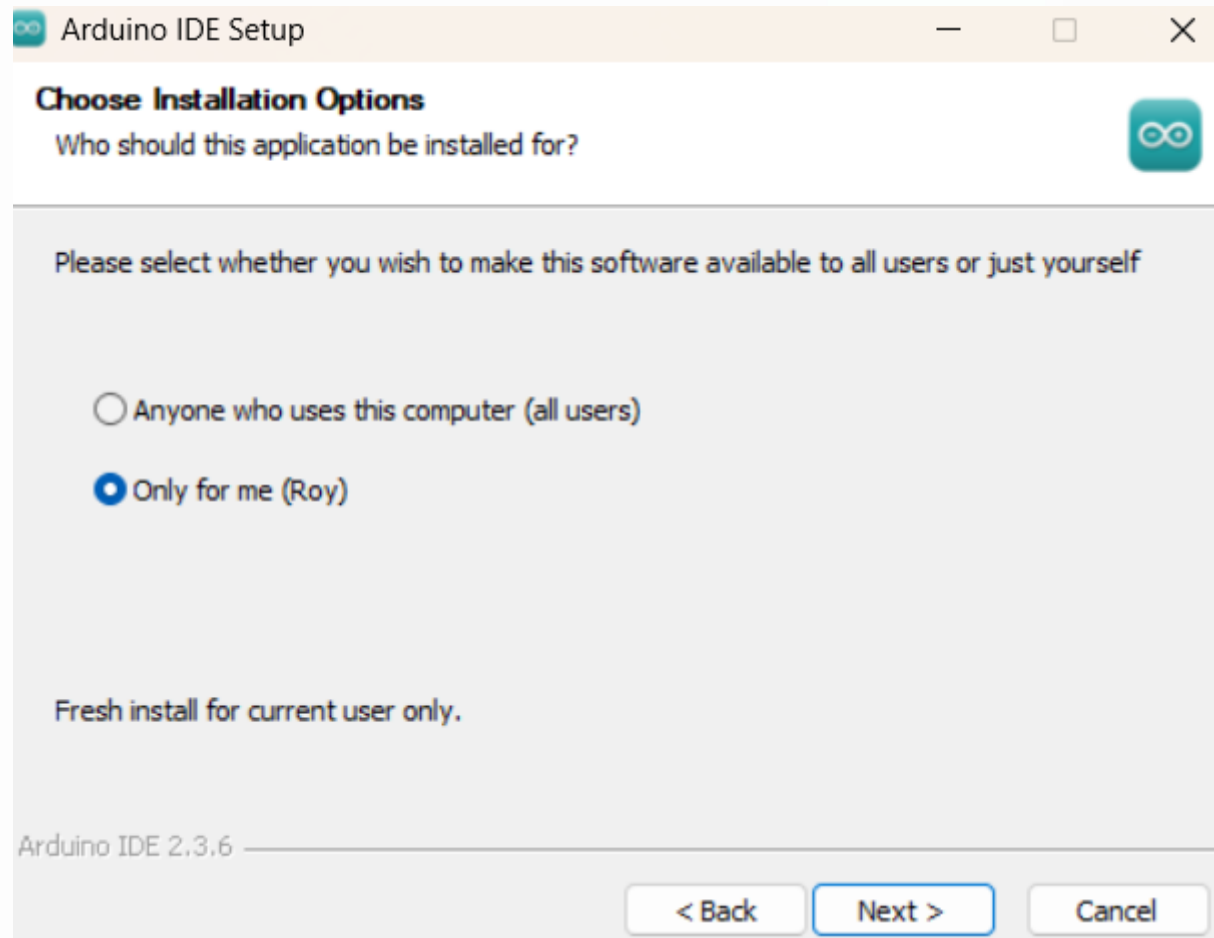
**Nightly Builds**
Download a preview of the incoming release with the most updated features and bugfixes.

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

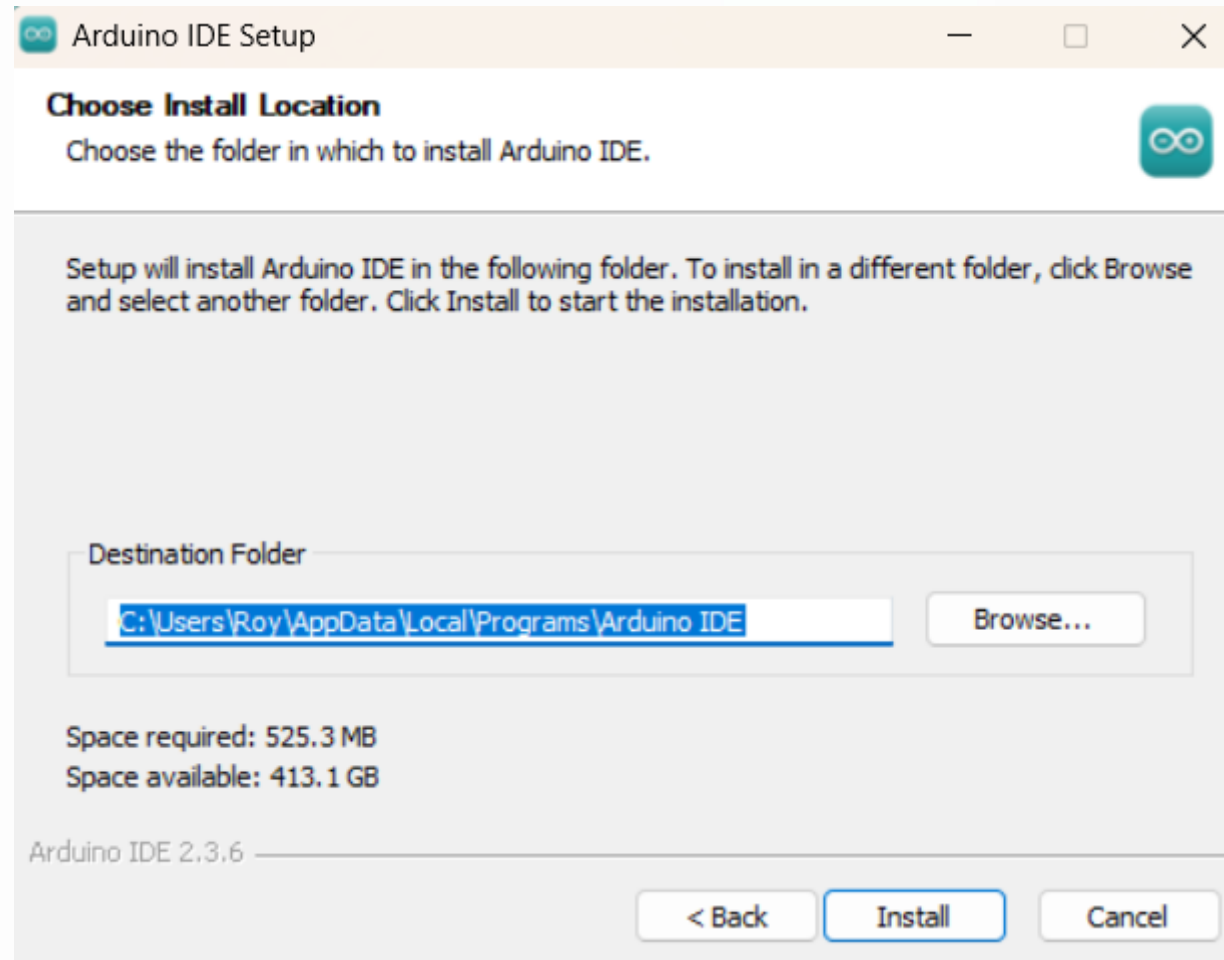
How to Setup Arduino IDE on your Local System



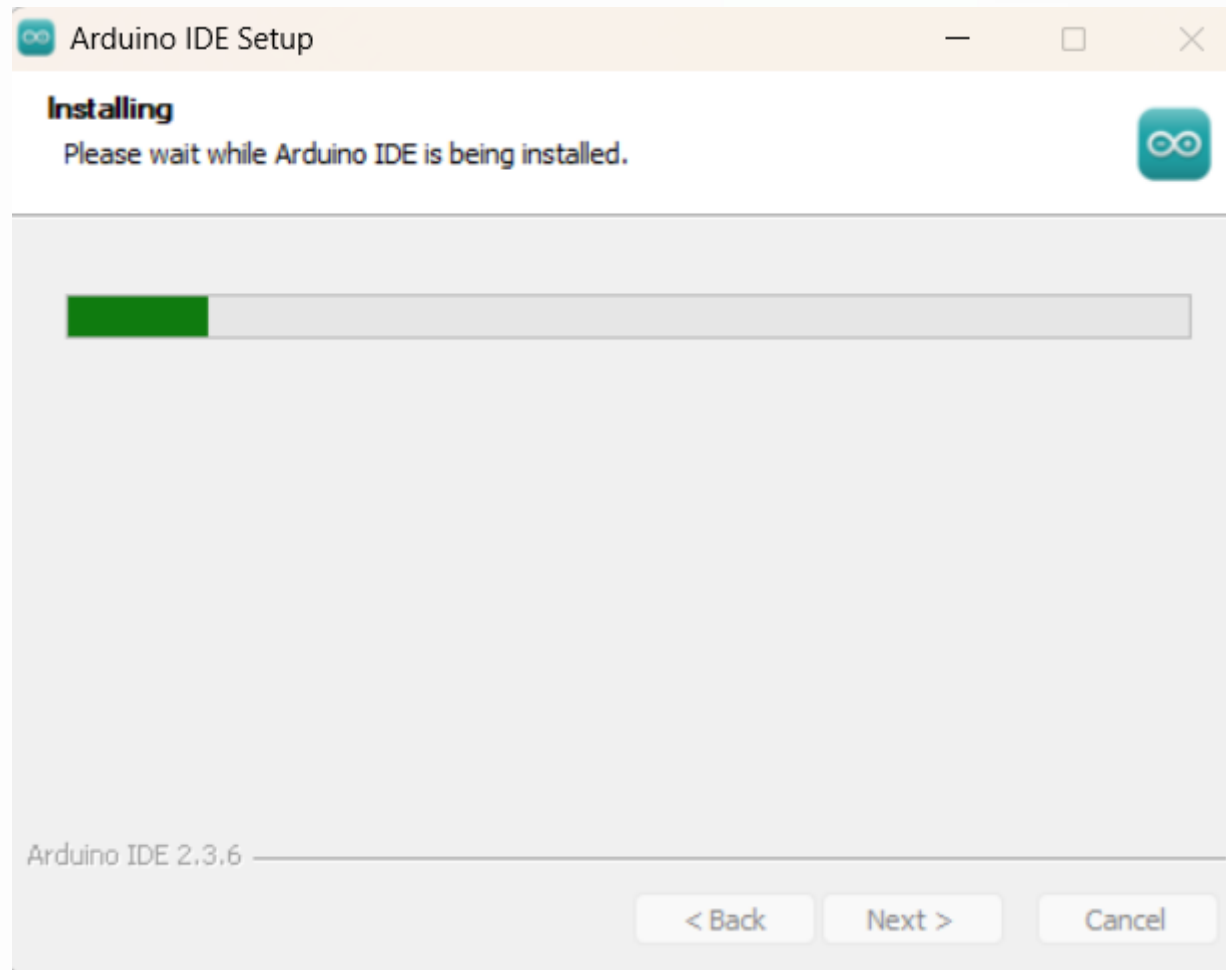
How to Setup Arduino IDE on your Local System



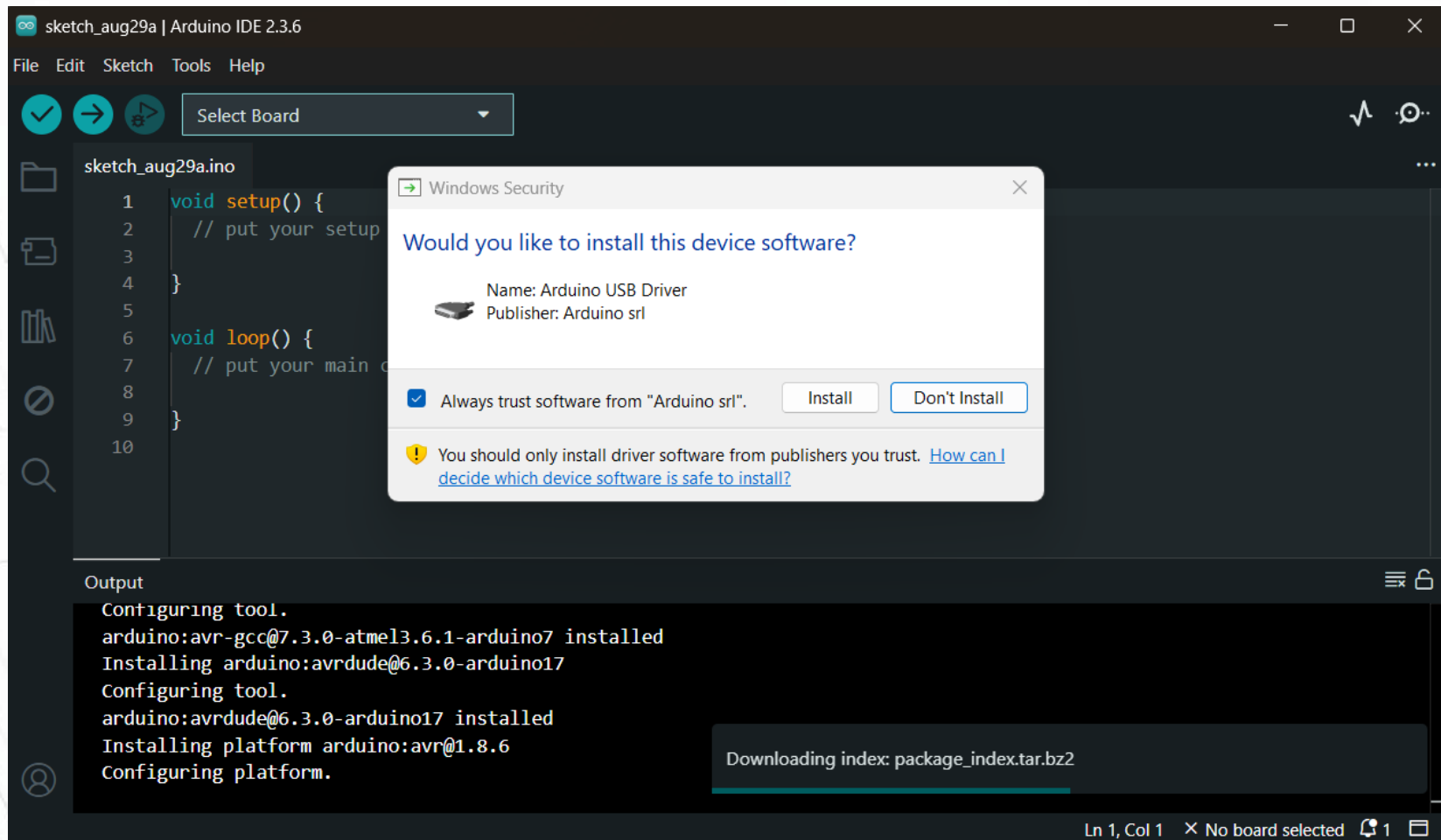
How to Setup Arduino IDE on your Local System



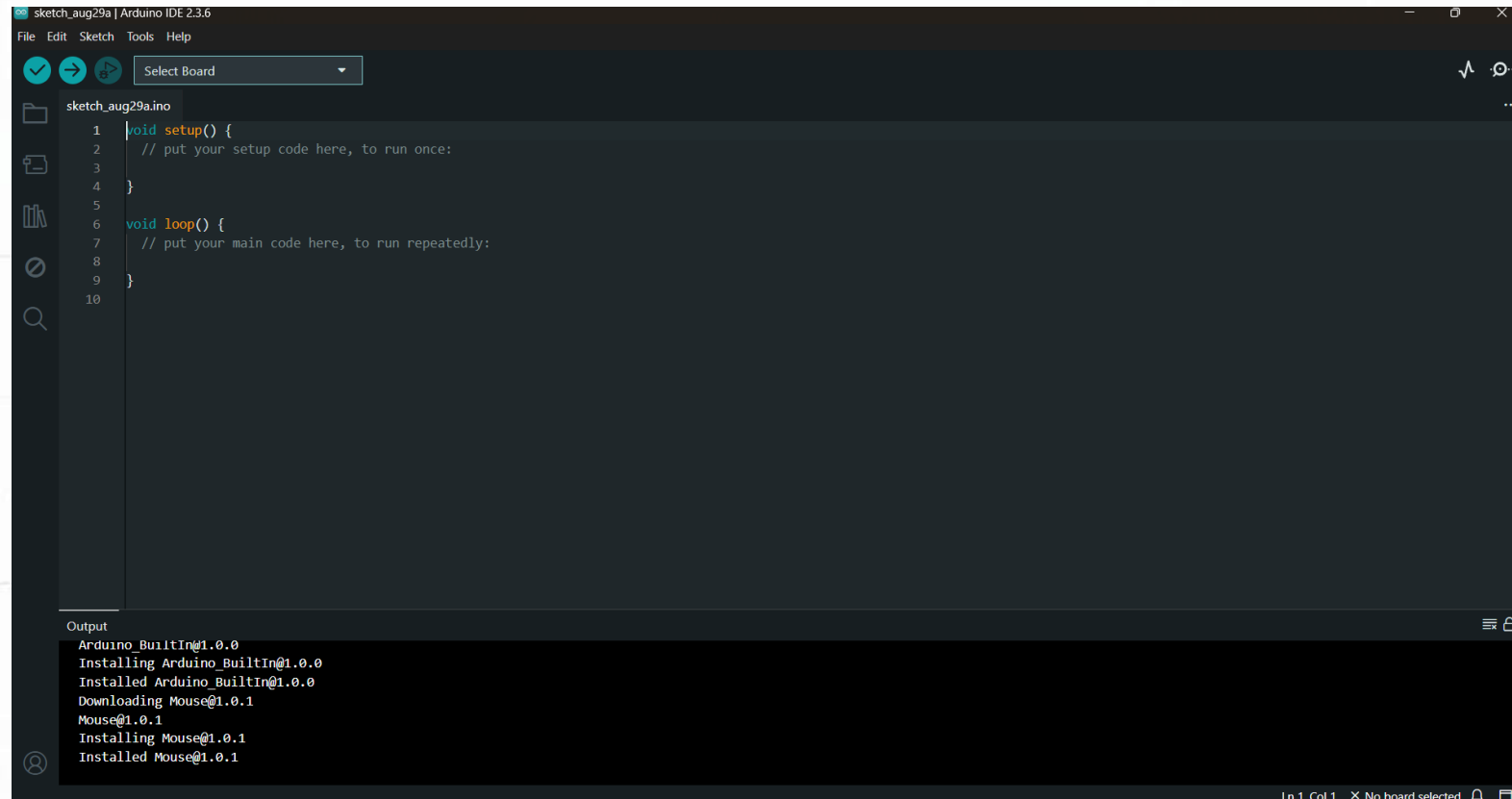
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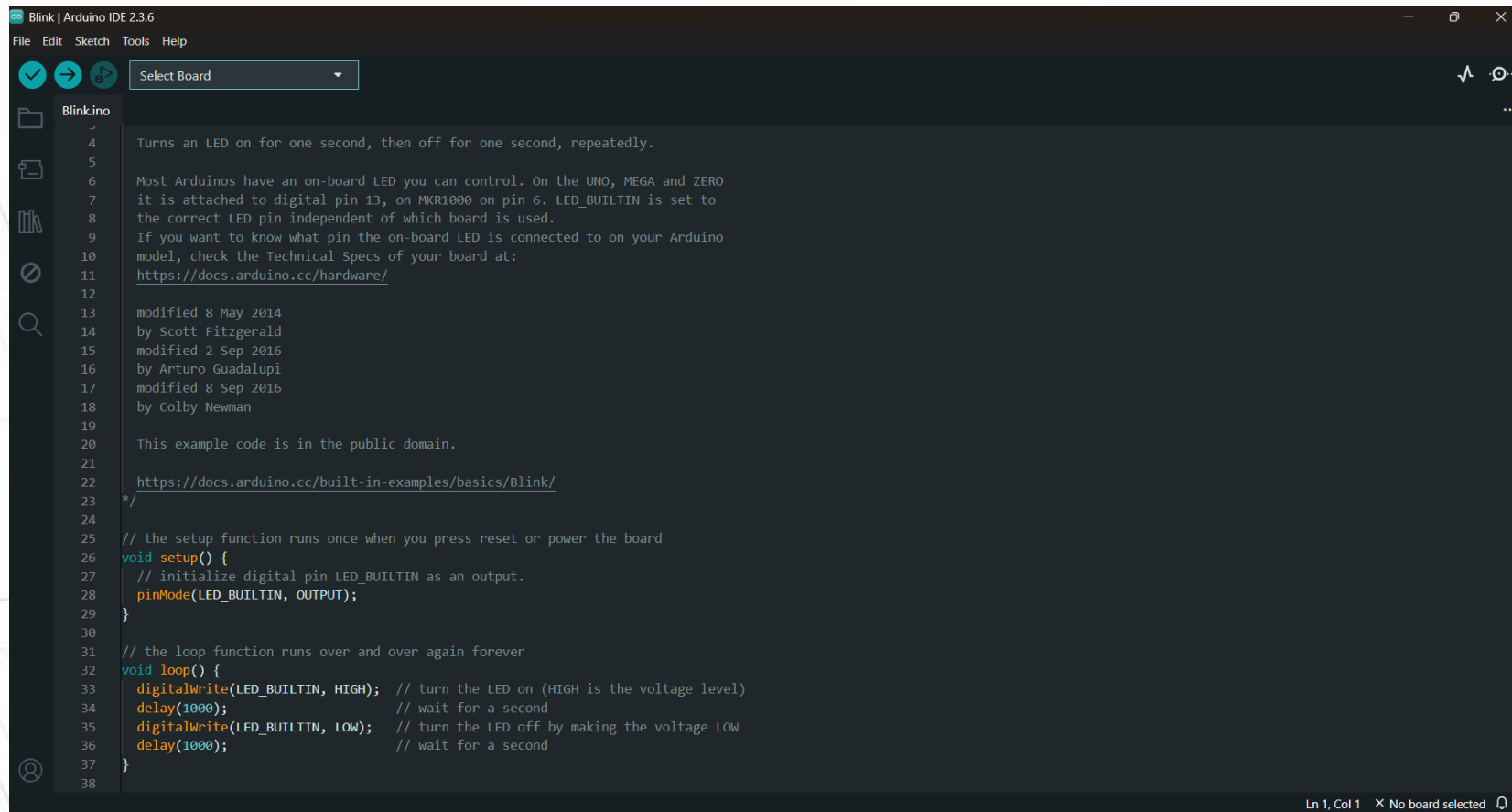
How to Setup Arduino on your Local System



How to Setup Arduino on your Local System



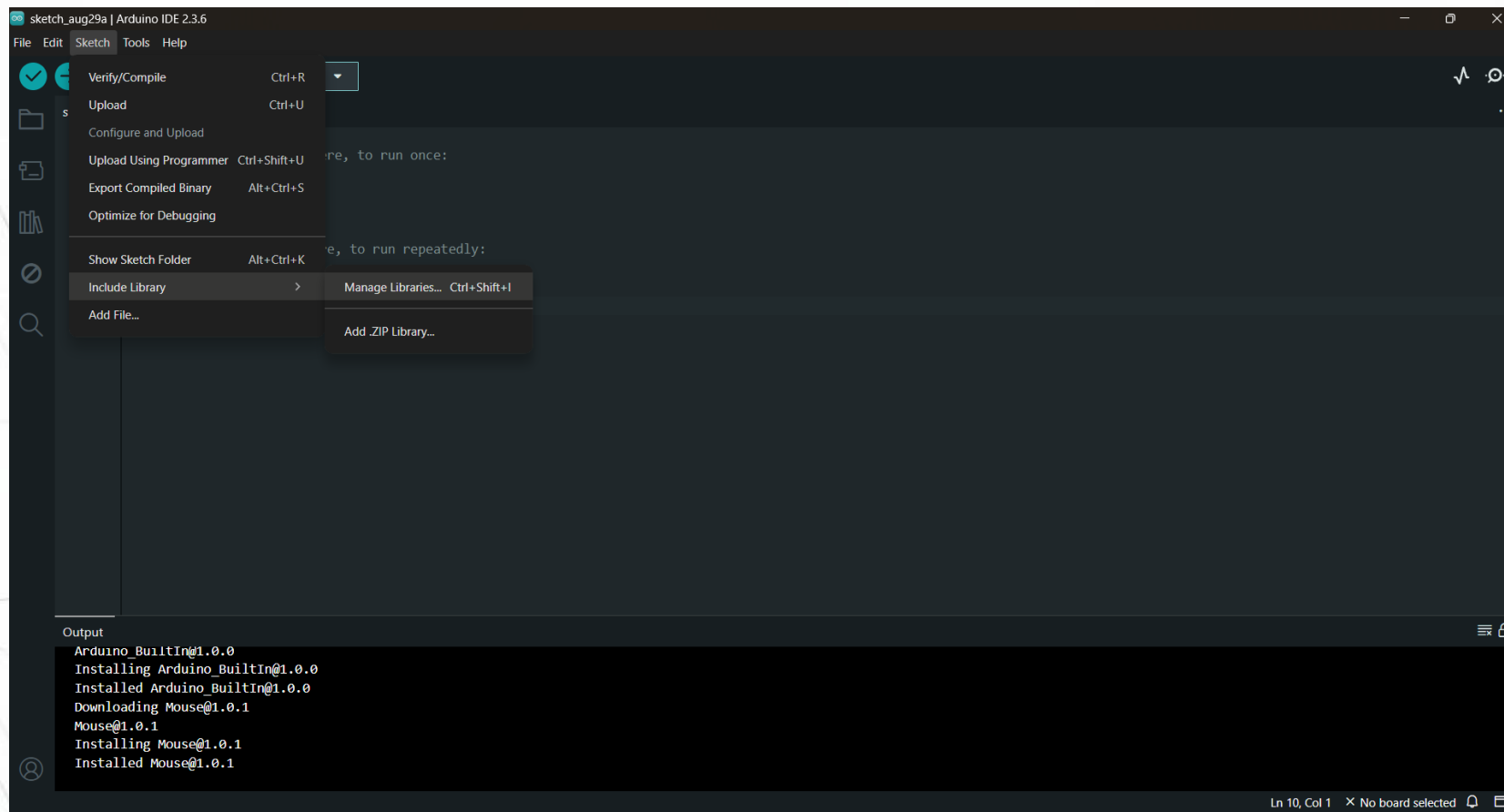
Blink Code Example



The screenshot shows the Arduino IDE 2.3.6 interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. Below the menu is a toolbar with icons for checking, running, and uploading code, along with a 'Select Board' dropdown menu. The left sidebar shows a file explorer with 'Blink.ino' selected. The main editor area displays the Blink code example, which includes a comment describing the code's function, a list of authors and their modification dates, and the actual C++ code for the setup and loop functions. The code uses the digitalWrite and delay functions to toggle an LED on and off every second. The status bar at the bottom indicates 'Ln 1, Col 1' and 'No board selected'.

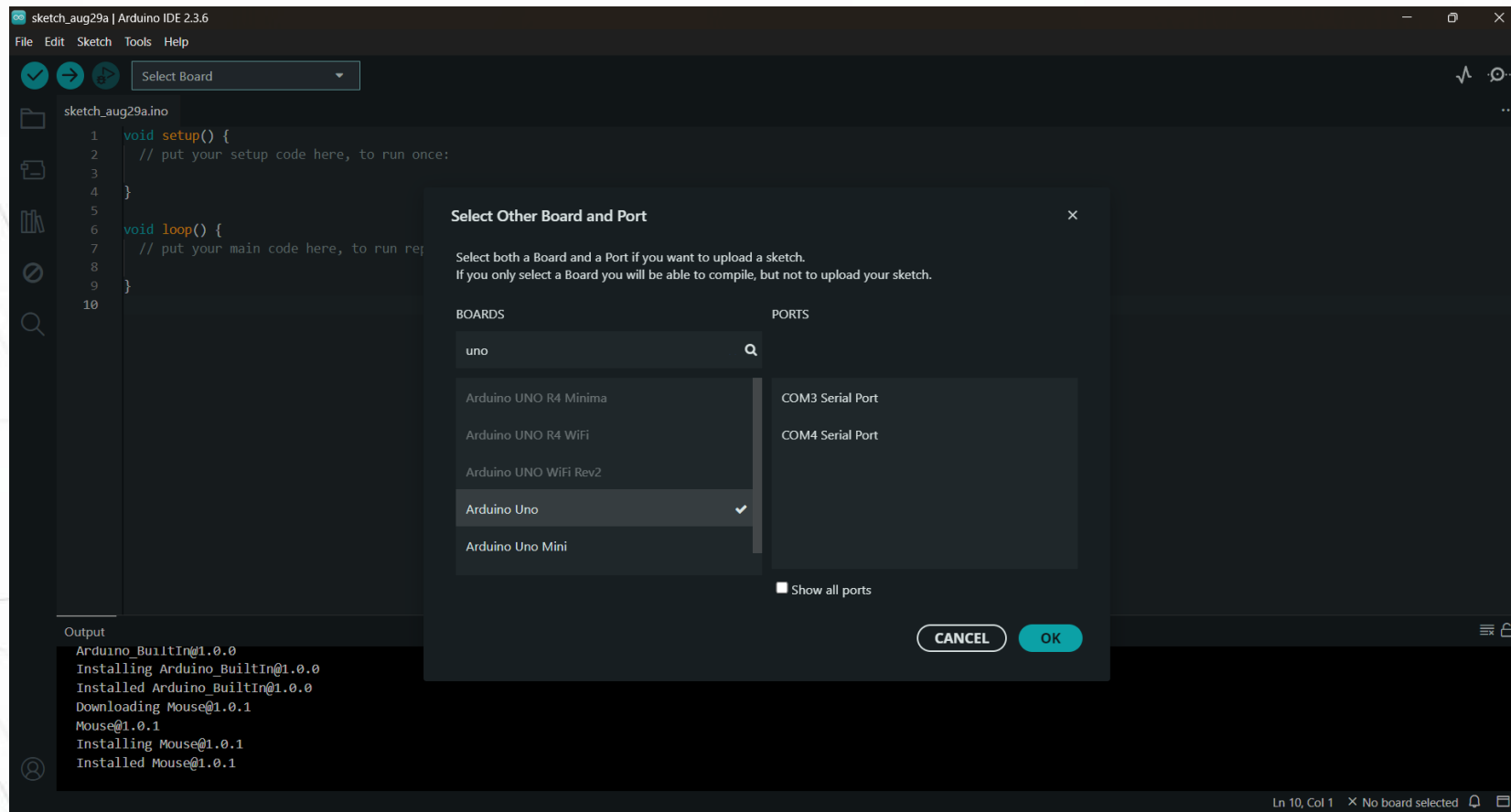
```
4 Turns an LED on for one second, then off for one second, repeatedly.
5
6 Most Arduinos have an on-board LED you can control. On the UNO, MEGA and ZERO
7 it is attached to digital pin 13, on MKR1000 on pin 6. LED_BUILTIN is set to
8 the correct LED pin independent of which board is used.
9 If you want to know what pin the on-board LED is connected to on your Arduino
10 model, check the Technical Specs of your board at:
11 https://docs.arduino.cc/hardware/
12
13 modified 8 May 2014
14 by Scott Fitzgerald
15 modified 2 Sep 2016
16 by Arturo Guadalupi
17 modified 8 Sep 2016
18 by Colby Newman
19
20 This example code is in the public domain.
21
22 https://docs.arduino.cc/built-in-examples/basics/Blink/
23 */
24
25 // the setup function runs once when you press reset or power the board
26 void setup() {
27   // initialize digital pin LED_BUILTIN as an output.
28   pinMode(LED_BUILTIN, OUTPUT);
29 }
30
31 // the loop function runs over and over again forever
32 void loop() {
33   digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
34   delay(1000); // wait for a second
35   digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW
36   delay(1000); // wait for a second
37 }
38
```

Library Management



Upload to Arduino

Select Board and Port Number



Reference

- <https://www.watelectronics.com/classification-of-embedded-systems/>
- Jiangxi University of Science & Technology
- My UG Prof: Dr. Ata Jahangir Moshayedi (drajm.com)



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Thank You

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